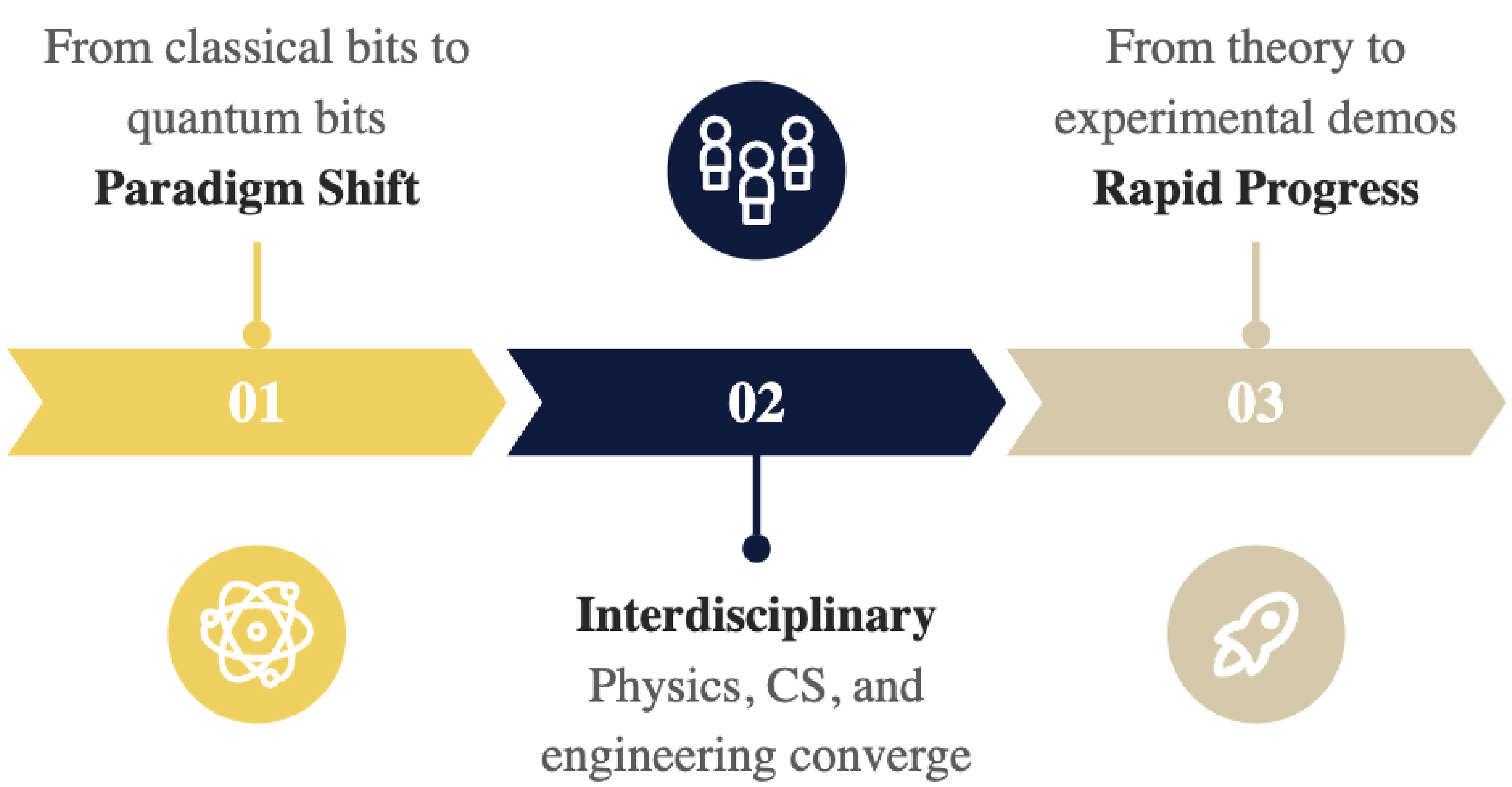
# The Evolution of Quantum Computing

Tracing the journey from abstract theory and seminal ideas to tangible hardware and modern breakthroughs.



# Theoretical Foundations and Origins

Exploring the seminal concepts that established the possibility and framework for quantum computation.

01

**Feynman's Proposal**

Using quantum systems to simulate nature

02

**Deutsch's Model**

Universal quantum model and Turing machine

03

**Computational Framework**

Harnessing quantum mechanics for computation

# Breakthrough Algorithms in Quantum Computing

Key algorithms that demonstrated the computational power and practical relevance of quantum devices.

### Shor's Algorithm

Efficient factoring, threatening encryption

### Grover's Algorithm

Quadratic speedup for unstructured search

### Hardware Motivation

Driving the build of functional quantum devices

# Hardware Platforms and Approaches

Comparing the diverse physical implementations striving to build scalable and reliable quantum processors.

### Superconducting

Fast gate operations, widely used

### Trapped Ion

High coherence, precise control

### Photonic

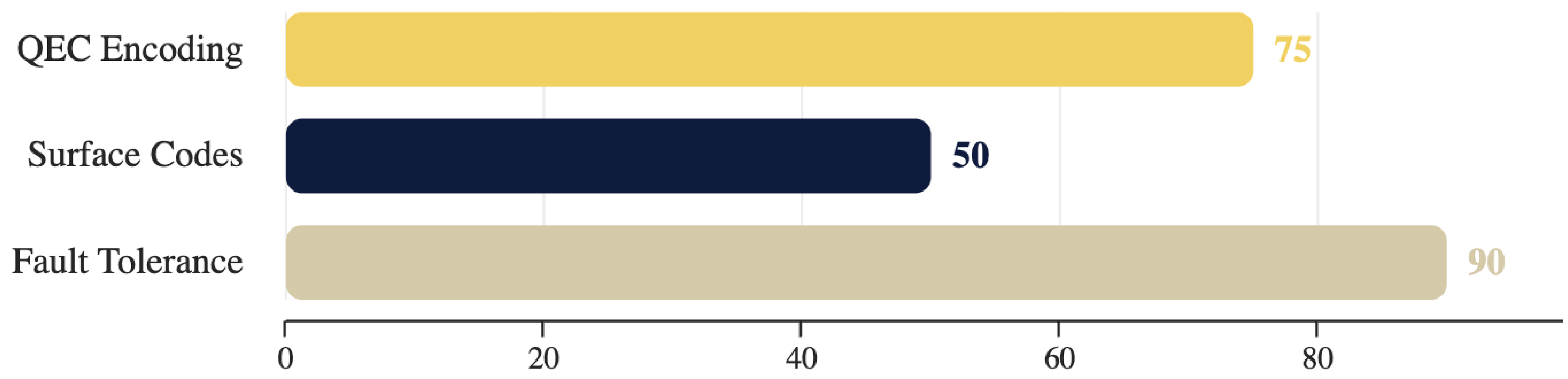
Room-temperature, networking advantages

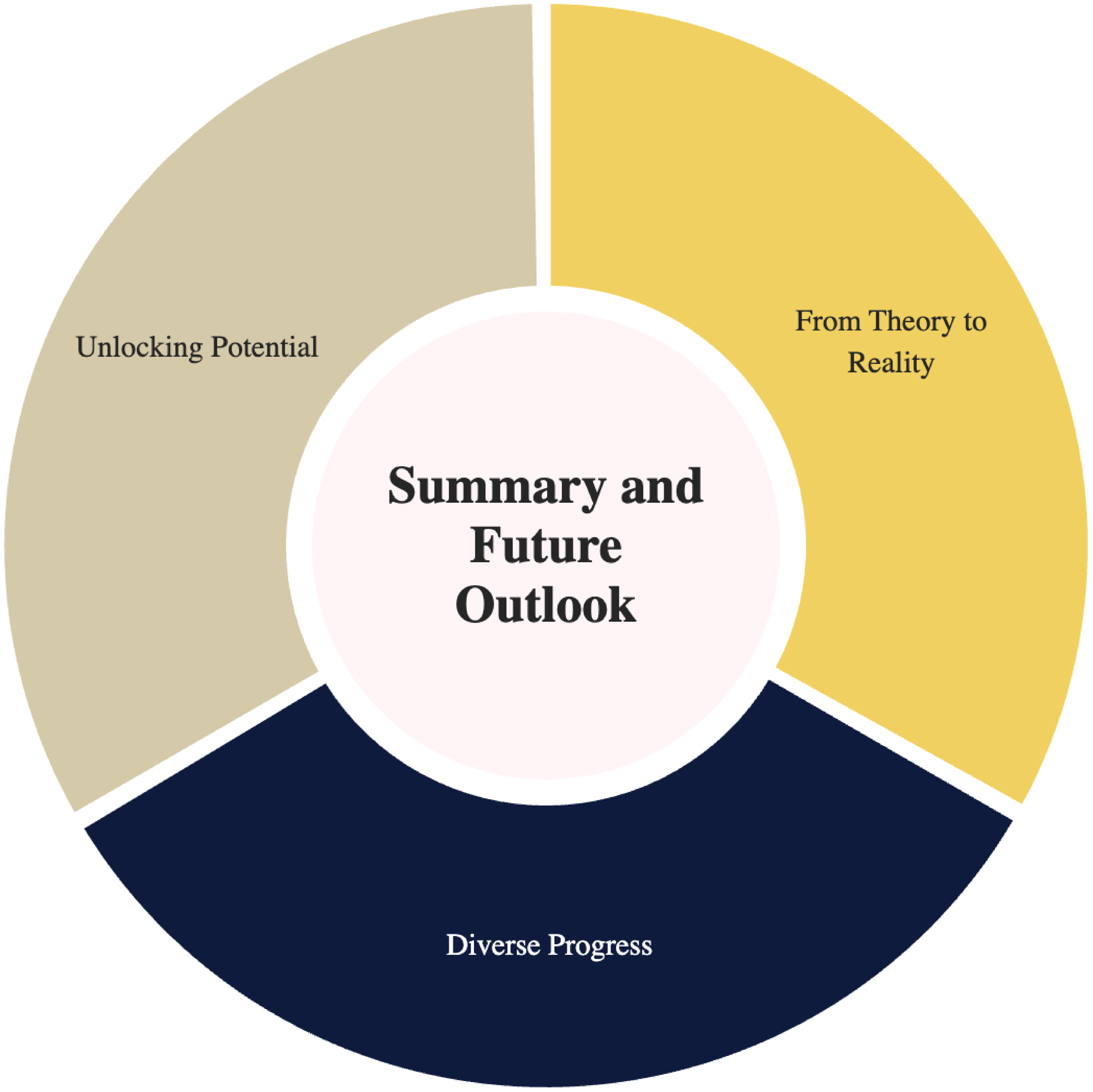
# The NISQ Era and Supremacy

Current landscape of noisy intermediate-scale quantum devices and milestone achievements in quantum advantage.

# Error Correction and Scaling

Addressing the critical challenge of noise and decoherence to realize fault-tolerant quantum computing.




Unlocking Potential
From Theory to Reality
Summary and Future Outlook
Diverse Progress